OH, $v = 2$

Hydroxy, X $^2\Pi_i$, $v = 2$

Species tag	017509
Version	1*
Date of Entry	Feb. 2023
Contributor	H. S. P. Müller

Details on the fit are given in
(1) A. Baudry, K. T. Wong, S. Etoka, A. M. S. Richards, H. S. P. Müller, F. Herpin, T. Danilovich, M. Gray, S. Wallström, D. Gobrecht, T. Khouri, L. Decin, C. A. Gottlieb, K. M. Menten, W. Homan, T. J. Millar, M. Montarges, B. Pimpanuwat, J. M. C. Plane, and P. Kervella, 2023, *Astron. Astrophys.* **674**, Art. No. A125.
The parameter set and the line list are largely based on the extensive isotopic invariant fit by
(2) B. J. Drouin, 2013, *J. Phys. Chem. A* **117**, 10076,
which, in turn, was the basis for the entries in the JPL catalog. The files were retrieved from the JPL catalog archive. We thank B. J. Drouin for the availability of the files. Two poorly determined parameters were omitted, and high-J Λ-doubling transitions were added from astronomical observations in (1) and in
(3) T. Khouri, L. Velilla-Prieto, E. De Beck, W. H. T. Vlemmings, H. Olofsson, B. Lankhaar, J. H. Black, and A. Baudry, 2019, *Astron. Astrophys.* **623**, Art. No. L1.
These data show systematic deviations with respect to the calculation in the JPL catalog.
Please note: The quantum numbers in the CDMS entry are Hund's case (b) quanta as always. They are N, Λ with parity, v, $J + 0.5$, and F. $N + 0.5 = J$ refers to $\Pi_{3/2}$ throughout, and $N - 0.5 = J$ to the higher lying $\Pi_{1/2}$. Thus, Hund's case (b) quanta are reasonable throughout.
The very extensive data set is described with references in (2). We provide here references for the Λ-doubling transitions and the various types of rotational and spin-flip transitions in $v = 2$. Λ-doubling transitions with microwave accuracy were reported by
(4) W. L. Meerts, J. P. Bekooy, and A. Dymanus, 1979, *Mol. Phys.* **37**, 425.
This publication involves measurements up to $N = 2$ in $\Pi_{3/2}$ and $\Pi_{1/2}$.
Rotational transition frequencies with microwave accuracy were reported by (2) and by
(5) T. D. Varberg and K. M. Evenson, 1993, *J. Mol. Spectrosc.* **157**, 55.
Rotational data with good FIR accuracies up to $N = 10$ were taken from
(6) M. A. Martin-Drumel, S. Eliet, O. Pirali, M. Guinet, F. Hindle, G. Mouret, and A. Cuisset, 2012, *Chem. Phys. Lett.* **550**, 8.
There are additional rovibrational lines in the data set which include some information on the Λ-doubling at higher rotational levels, but it is not clear if this information is sufficient. Therefore, Λ-doubling transitions above 70 GHz should be viewed with some caution, above 100 GHz with considerable caution. Some caution is also advised for rotational transitions and transitions involving spin-flip if the calculated uncertainties exceed 1 MHz.
The partition function was evaluated by summing over all states up to $F = 50$ and $v = 9$. While the partition function is essentially converged at 3000 K, model deficiencies may make it advisable to view values at 2000 K and higher with some caution.
The $v = 1$ dipole moment was determined in
(7) K. I. Peterson, G. T. Fraser, and W. Klemperer, 1984, *Can. J. Phys.* **62**, 1502.

Lines Listed	722
Frequency / GHz	<35000
Max. J	21
log STR0	-30.0
log STR1	-30.0
Isotope Corr.	
Egy / cm^{-1}	0.0
μ_a / D	1.665
μ_b / D	
μ_c / D	
A / MHz	
B / MHz	513318.51
C / MHz	
Q(3000.)	1124.1256
Q(2500.)	869.7149
Q(2000.)	648.8509
Q(1500.)	458.1936
Q(1000.)	291.9613
Q(500.0)	139.9991
Q(300.0)	81.4940
Q(225.0)	60.2961
Q(150.0)	40.1443
Q(75.00)	22.7535
Q(37.50)	17.0337
Q(18.75)	16.0037
Q(9.375)	15.9294
Q(5.000)	15.8680
Q(2.725)	15.7594
detected in ISM/CSM	probably not

